

Rational Integral Operators with Distinguished Fibres: A Linear-Algebraic Characterization of the BBP Numerator and a General Classification Problem

Dean Kulik

QuHarmonics Research Group · ORCID 0009-0003-3128-8828

NEXUS / A-Mark9 Framework · August 2026 · v1.0

Abstract

The Bailey–Borwein–Plouffe formula is usually presented as a digit-extraction algorithm for π . We instead treat its numerator as a vector and characterize it algebraically. Fixing the denominator $P(t) = 16 - t^8$, the map sending an 8-coefficient numerator to the coordinate vector of $16 \int_0^1 N(t)/P(t) dt$ in a fixed generating family of transcendentals is a linear operator Φ over the number field $\mathbb{Q}(\sqrt{2})$. We prove Φ has rank 7 (hence a one-dimensional kernel), so the preimage of the π -line is a two-dimensional space. A single linear functional — the coefficient sum Σ , which by a boundary-evaluation identity equals $N(1)$ — cuts this two-dimensional fibre to a unique line, and the classical BBP numerator spans it. Every load-bearing quantity is quoted from live symbolic and 60-digit numerical computation, with a reproducible verification script. We separate this proved core cleanly from its generalization: the phenomenon (a low-dimensional distinguished fibre selected by a natural boundary functional) is posed as an operator-classification problem over square-free P , which the present result establishes as non-empty.

o. Computational foundation and reading order

This paper is arranged as a dependency chain: each result is the input to the next, and the whole chain rests on one constructed object, the coefficient matrix M . All symbolic factorization and integration were performed with SymPy; every matrix entry was independently certified against a 60-digit numerical quadrature of $16 t^p/P(t)$. The maximum entry error was 1.94×10^{-61} , certifying M to more than fifty digits. All theorems below are conditional only on the correctness of these certified entries; given M , every remaining step is finite linear algebra over a number field and requires no transcendence theory. A standalone verification script reproduces the entire chain (Appendix C).

The denominator factors over $\mathbb{Q}$ as

$$16 - t^8 = -(t^2 - 2)(t^2 + 2)(t^2 - 2t + 2)(t^2 + 2t + 2)$$

giving four quadratic factors: one with real roots ($\pm\sqrt{2}$), three with complex roots. The eight roots lie on the circle of radius $\sqrt{2}$ at the eighth roots of 16 — a single geometric fact from which both the base (16) and the shell count (8) are read, since $16 = (\sqrt{2})^8$.

1. The formal operator Φ and its matrix

Let $W = \mathcal{Q}$ be a free module with formal basis indexed by the seven transcendentals that appear when the atomic partial fractions of $1/P$ are integrated over $[0,1]$:

$$F = (f_{\pi}, f_{\log 2}, f_{\log 3}, f_{\log 5}, f_{\log(1+\sqrt{2})}, f_{\operatorname{atan}(1/\sqrt{2})}, f_{\operatorname{atan} 2})$$

Define $\Phi: \mathcal{Q}^8 \rightarrow W$ by $\Phi(\mathbf{n}) = M^T \mathbf{n}$, where the row p of M records the coordinates of $16 \int_0^1 t^p/P(t) dt$ in the family F . The physical operator is the composition $L = \operatorname{ev} \circ \Phi$, where ev sends each formal symbol to its real value. Separating Φ (formal, over $\mathbb{Q}(\sqrt{2})$) from ev (evaluation into $\mathbb{R}$) is deliberate: it keeps the entire rank argument inside a number field and removes any dependence on linear independence of the transcendentals themselves.

The certified matrix M

Written with $s = \sqrt{2}$, the matrix $M \in \operatorname{Mat}_{8 \times 7}(\mathbb{Q}(\sqrt{2}))$ is:

p	f_{π}	$f_{\log 2}$	$f_{\log 3}$	$f_{\log 5}$	$f_{\log(1+\sqrt{2})}$	$f_{\operatorname{atan}(1/\sqrt{2})}$	$f_{\operatorname{atan} 2}$
0	0	0	0	1/8	$s/4$	$s/4$	1/4
1	1/4	0	1/4	0	0	0	-1/2
2	0	0	0	-1/4	$s/2$	$-s/2$	1/2
3	0	0	-1/2	1/2	0	0	0
4	0	0	0	-1/2	s	s	-1
5	-1	0	1	0	0	0	2
6	0	0	0	1	$2s$	$-2s$	-2

p	f _π	f _{log2}	f _{log3}	f _{log5}	f _{log(1+√2)}	f _{atan(1/√2)}	f _{atan2}
7	0	8	-2	-2	0	0	0

Numerical certification (row error vs 60-digit quadrature):

p=0: |recon - l| = 0.0 p=4: |recon - l| = 1.94e-61
p=1: |recon - l| = 0.0 p=5: |recon - l| = 1.36e-61
p=2: |recon - l| = 1.17e-61 p=6: |recon - l| = 5.83e-62
p=3: |recon - l| = 0.0 p=7: |recon - l| = 7.78e-62
max entry error = 1.94e-61

2. Theorem 1 — rank(M) = 7

Theorem 1. Given the certified matrix M, $\text{rank}_{\mathbb{Q}(\sqrt{2})}(M) = 7$.

Proof. Of the eight possible 7×7 minors (each obtained by omitting one row), seven are nonzero. Their exact values are:

omit row 0: det = 16 omit row 4: det = -4
omit row 1: det = 16 omit row 5: det = 4
omit row 2: det = -8 omit row 6: det = 2
omit row 3: det = 16 omit row 7: det = 0

Each nonzero value lies in $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$, so at least one 7×7 minor is nonzero and $\text{rank}(M) = 7$. ■

Corollary 2. $\dim_{\mathbb{Q}(\sqrt{2})} \ker(\Phi) = 1$. By rank-nullity on $M^T : \mathbb{Q}^8 \rightarrow W$, the nullity is $8 - 7 = 1$. Explicitly,

$\ker(\Phi) = \text{span}\{z\}$, $z = [-8, 8, 4, 8, 2, 2, -1, 0]$, $M^T z = 0$.

3. The π-fibre

Proposition 3. $\Phi^{-1}(\text{span}\{f_{\pi}\}) = \text{span}\{v_1\} \oplus \ker(\Phi)$, a two-dimensional space.

A particular solution is $v_1 = [-4, 0, 0, 2, 1, 1, 0, 0]$, which satisfies $M^T v_1 = [-1, 0, 0, 0, 0, 0, 0, 0] = -f_{\pi}$. Its negation is exactly the classical BBP numerator $4 - 2t^3 - t^4 - t^5$, and $L(-v_1) = \pi$.

Proof. If $\Phi(v) = c \cdot f_{\pi}$ then $\Phi(v + c \cdot v_1) = c \cdot f_{\pi} + c \cdot (-f_{\pi}) = 0$, so $v + c \cdot v_1 \in \ker(\Phi)$. Hence $v \in \text{span}\{v_1\} + \ker(\Phi)$; the sum is direct because $v_1 \notin \ker(\Phi)$. ■

Sign remark. Proposition 3 uses $\Phi(v_1) = -f_{\pi}$, not $+f_{\pi}$. The cancellation $\Phi(v + c \cdot v_1) = c \cdot f_{\pi} + c \cdot \Phi(v_1)$ requires this sign; the BBP numerator itself is $-v_1$, and produces $+\pi$. The convention is carried consistently below.

4. The boundary functional and the Boundary Evaluation Identity

Let $\Sigma : \mathbb{Q}^8 \rightarrow \mathbb{Q}$ be the coefficient-sum functional $\Sigma(n) = \sum n_p = N(1)$. The following identity ties Σ to the annihilation structure of the operator.

Boundary Evaluation Identity. Let $D(t) = b + s \cdot t^n$ with $s = \pm 1$ and $b + s \neq 0$, and let $\deg N < n$. Put $A = \gcd(N, D)$ and $M = N/A$ in $\mathbb{Q}[t]$. Then

$N(1) = \sum n_p = A(1) \cdot M(1)$, with $A(1) \mid D(1) = b + s$, $A(1) \neq 0$.
Consequently $\sum n_p = 0 \Leftrightarrow M(1) = 0$.

Proof. $N = A \cdot M$ in $\mathbb{Q}[t]$; evaluate at $t = 1$. Since $A \mid D$, $A(1) \mid D(1) = b + s \neq 0$, so $A(1) \neq 0$. ■

For $P(t) = 16 - t^8$ we have $P(1) = 15 = 3 \cdot 5$, the boundary values of the two quadratics the BBP numerator annihilates. The selection condition is therefore written in coordinates the annihilation itself leaves behind. The identity was tested across seven constructions on five distinct denominators:

$D(t)$	coefficient vector n	value	$\sum n_p$	$A(1)$	$M(1)$	$A(1) \cdot M(1)$	ID
$16 - t^8$	$[-4, 0, 0, 2, 1, 1, 0, 0]$	π	0	15	0	0	✓
$16 - t^8$	$[0, -8, -4, -4, 0, 0, 1, 0]$	π	-15	15	-1	-15	✓
$4 + t^4$	$[8, 8, 4, 0]$	π	20	5	4	20	✓
$1 + t^2$	$[4, 0]$	π	4	1	4	4	✓
$1 + t^4$	$[0, 8, 0, 0]$	π	8	1	8	8	✓
$1 + t^8$	$[0, 0, 0, 16, 0, 0, 0, 0]$	π	16	1	16	16	✓
$64 - t^6$	$[16, -24, -8, -6, 1, 0]$	0	-21	7	-3	-21	✓

The $64 - t^6$ row is the *zero relation* — its value is exactly 0, which lies in $\mathbb{Q}\pi$, so the identity applies there as well. Lemma 4. $\Sigma(v_1) = 0$ and $\Sigma(z) = 15$ (direct computation).

5. Theorem 5 — the BBP numerator is uniquely selected

Theorem 5. $(\text{span}\{v_1\} \oplus \ker(\Phi)) \cap \ker(\Sigma) = \text{span}\{v_1\}$.

Proof. Every element of the π -fibre has the form $v = a \cdot v_1 + b \cdot z$. By Lemma 4, $\Sigma(v) = a \cdot 0 + b \cdot 15 = 15b$, so $\Sigma(v) = 0 \Leftrightarrow b = 0 \Leftrightarrow v = a \cdot v_1$. ■

Geometrically: the kernel direction z is *transverse* to the hyperplane $\ker(\Sigma)$, because Σ vanishes on v_1 but not on z . The intersection is therefore exactly the remaining line. **Corollary 6.** The classical BBP numerator spans the unique one-dimensional intersection of the π -fibre with the DC-blocking hyperplane $\Sigma = 0$.

Scope. Theorem 5 characterizes the BBP numerator within the operator L for $P(t) = 16 - t^8$. It does not assert that every digit-extraction formula reduces to this operator, nor that this is the only denominator with the property. Those are the subject of §6.

6. The general classification problem

The proved core exhibits a specific structure: an integral operator whose preimage of a distinguished constant is low-dimensional, and a single boundary-derived functional that cuts that preimage to a unique ray. This structure is not obviously particular to base 16. We state the general object and the open question it raises.

Let $P \in \mathbb{Q}[t]$ be square-free of degree n , and let W_P be a free $\mathbb{Q}$ -module with basis indexed by the formal symbols appearing in the partial-fraction decomposition of $1/P$. Define

$$\Phi_P : \mathbb{Q}[t]_{<n} \rightarrow W_P, \quad \Phi_P(N) = [\text{coordinate vector of } \int_0^1 N/P].$$

Classification problem. For which square-free P does there exist a linear functional Σ on $\mathbb{Q}[t]_{<n}$ arising from a boundary evaluation identity, and a distinguished formal symbol f_c , such that $\Phi_P^{-1}(\text{span}\{f_c\}) \cap \ker(\Sigma)$ is one-dimensional?

The present result establishes that this set of denominators is non-empty: $P(t) = 16 - t^8$ is a member, with $c = \pi$. The natural next data point is a second, structurally unrelated member — a denominator whose distinguished constant is not π (for instance a target of e or of a logarithm), which would convert the phenomenon from an isolated instance into a genuine class. Two independent conditions govern membership and are worth isolating in future work: (i) that the preimage of the distinguished constant be exactly two-dimensional (so a single functional suffices to cut it), and (ii) that the boundary functional be transverse to the kernel. The dimension count of §2–3 shows both hold for $16 - t^8$; neither is automatic for arbitrary P .

7. Dependency graph and node grading

Each node's status is graded so that a challenge to any single node exposes exactly the downstream results it affects.

```
[Factorization of P] --sympy.factor
|
[Partial fractions] --sympy.apart
|
[Atomic integrals] --sympy.integrate
|
[Matrix M] --assembled, certified to 55+ digits vs quadrature
|
+--> [rank(M) = 7] --7 nonzero 7x7 minors      (THEOREM 1)
|  |
|  [dim ker(Phi) = 1] --rank-nullity      (COROLLARY 2)
|  |
|  [ker(Phi) = span{z}] --explicit vector
|
+--> [particular solution v1]
|
[pi-fibre = span{v1} (+) ker(Phi)]      (PROP 3)
|
[Boundary Evaluation Identity] --N(1)=A(1)M(1) (THEOREM)
|
[Sigma(v1)=0, Sigma(z)=15]      (LEMMA 4)
|
[pi-fibre n ker(Sigma) = span{v1}]      (THEOREM 5)
```

Node	Grade	Verification
Factorization of P	THEOREM	sympy.factor + expand check

Node	Grade	Verification
Matrix M	CONSTRUCTION	assembled; certified vs 60-digit quadrature
$\text{rank}(M) = 7$	THEOREM	7 nonzero 7×7 minors over $\mathbb{Q}(\sqrt{2})$
$\dim \ker(\Phi) = 1$	THEOREM	rank-nullity
$\ker(\Phi) = \text{span}\{z\}$	THEOREM	$M^T z = 0$, explicit
$\pi\text{-fibre} = \text{span}\{v_1\} \oplus \ker$	THEOREM	standard linear algebra (Prop 3)
Boundary Evaluation Identity	THEOREM	polynomial identity, 7 constructions
$\Sigma(v_1)=0, \Sigma(z)=15$	VERIFIED	direct computation (Lemma 4)
BBP uniquely selected	THEOREM	Prop 3 + Lemma 4 (Theorem 5)
General classification	OPEN	non-empty; membership conditions stated

8. Open problems

- **Second member.** Exhibit a square-free P , structurally unrelated to $16 - t^8$ and with a distinguished constant other than π , satisfying the classification condition. This is the single result that upgrades the phenomenon from instance to class.
- **Dimension criterion.** Characterize which P give a two-dimensional distinguished fibre. The dimension equals $\dim(\text{image of the distinguished constant}) + \dim(\text{kernel})$; state when the kernel is exactly one-dimensional in terms of the pole structure of P .
- **Transversality.** Give conditions on P under which the boundary functional Σ is transverse to $\ker(\Phi)$, so a single constraint suffices for uniqueness.
- **Weight sequences.** The construction fixes the harmonic weights implicit in $\int t^p/P$. Determine which of the load-bearing facts survive replacing $1/(j+1)$ by $1/(j+1)^k$ (di- and tri-logarithmic readouts).

9. Interpretation

The results of §§1–5 stand independently of everything in this section. What follows is a reading, not a consequence; it is offered because the algebra leaves a specific and narrow opening for it.

Two features of the proved core are worth naming structurally. First, the operator *adds nothing*: after annihilation the surviving numerator is $M(t) = 1 - t$, degree one, and the boundary identity reads the selection condition $\Sigma = 15 \cdot M(1)$ directly off the poles the numerator already killed. The selection and the annihilation are not two steps but one object read twice. Second, the extra dimension in the π -fibre is not a property of π . It is the count of coordinates the reader committed to: eight monomial taps sampling a seven-dimensional image, one redundant. The kernel vector z is that redundancy made explicit. Change the sampling — the basis, the tap count — and the redundant dimension moves, while the image does not.

This is the precise sense in which the incompleteness belongs to the reader and not to the object: the invariant π belongs to the operator, and any faithful realization of that operator exposes the same invariant,

independent of substrate. Whether any particular physical system realizes this operator is an empirical question this paper does not address and its mathematics does not require. The classification problem of §6 is the mathematically tractable residue of that larger question — it asks which operators have the structure, not which things in the world instantiate them.

Appendix A — The matrix M and atomic integrals

The eight atomic integrals (from partial fractions of $1/P$ over $\mathbb{Q}(\sqrt{2})$):

$$\begin{aligned}\int_0^1 t/(t^2-2) \, dt &= -(1/2) \log 2 \\ \int_0^1 1/(t^2-2) \, dt &= -(1/\sqrt{2}) \log(1+\sqrt{2}) \\ \int_0^1 t/(t^2+2) \, dt &= (1/2)(\log 3 - \log 2) \\ \int_0^1 1/(t^2+2) \, dt &= (1/\sqrt{2}) \operatorname{atan}(1/\sqrt{2}) \\ \int_0^1 t/(t^2-2t+2) \, dt &= -(1/2) \log 2 + \pi/4 \\ \int_0^1 1/(t^2-2t+2) \, dt &= \pi/4 \\ \int_0^1 t/(t^2+2t+2) \, dt &= -(1/2)\log 2 + (1/2)\log 5 - \operatorname{atan} 2 + \pi/4 \\ \int_0^1 1/(t^2+2t+2) \, dt &= \operatorname{atan} 2 - \pi/4\end{aligned}$$

Appendix B — Boundary Evaluation Identity (full statement)

Theorem. Let $D(t) = b + s \cdot t^n$ with $s = \pm 1$ and $b + s \neq 0$. Let $\deg N < n$. Set $A = \gcd(N, D)$ and $M = N/A$ in $\mathbb{Q}[t]$. Then (1) $N(1) = \sum n_p = A(1) \cdot M(1)$; (2) $A(1) \mid D(1) = b + s$; (3) $A(1) \neq 0$; (4) $\sum n_p = 0 \Leftrightarrow M(1) = 0$.

Proof. $N = A \cdot M$ in $\mathbb{Q}[t]$; evaluate at $t = 1$. ■

Appendix C — Standalone verification

The following reproduces the entire Layer-1 chain. Every printed value is quoted in the body of this paper.

```
import sympy as sp
from sympy import Rational as R, sqrt, Matrix
import mpmath; mpmath.mp.dps = 60
t = sp.Symbol('t'); P = 16 - t**8; s2 = sqrt(2)

# factorization
assert sp.expand(-(t**2-2)*(t**2+2)*(t**2-2*t+2)*(t**2+2*t+2) - P)==0

M = Matrix([
    [0,0,0,R(1,8),s2/4,s2/4,R(1,4)],
    [R(1,4),0,R(1,4),0,0,0,-R(1,2)],
    [0,0,0,-R(1,4),s2/2,-s2/2,R(1,2)],
    [0,0,-R(1,2),R(1,2),0,0,0],
    [0,0,0,-R(1,2),s2,s2,-1],
    [-1,0,1,0,0,0,2],
    [0,0,0,1,2*s2,-2*s2,-2],
    [0,8,-2,-2,0,0,0]])

# rank(M) = 7 via 7x7 minors
from itertools import combinations
nz = [sp.simplify(M[list(r),:].det()) for r in combinations(range(8),7)]
assert sum(1 for d in nz if d!=0) == 7

# kernel, particular solution, uniqueness
z = Matrix([-8,8,4,8,2,2,-1,0]); assert M.T*z == sp.zeros(7,1)
v1 = Matrix([-4,0,0,2,1,1,0,0])
assert list(M.T*v1) == [-1,0,0,0,0,0,0]
```



```
assert sum(v1)==0 and sum(z)==15 # Sigma(a v1 + b z) = 15b => unique at b=0
print('ALL CHECKS PASS')
```

Assuming the certified matrix entries, the strongest derived result is: for $P(t) = 16 - t^8$, the operator Φ has a two-dimensional π -fibre, and the coefficient-sum functional — arising from the Boundary Evaluation Identity — isolates a unique one-dimensional subspace, spanned by the classical BBP numerator. Everything beyond this is stated as interpretation or as the open classification problem.